

Defining Requirements

INTRODUCTION

Previous editions of the Nutrient Requirements of Dairy Cattle (e.g., NRC, 2001) reported requirements for various nutrients in dairy cattle without specifically defining what the term “nutrient requirement” actually meant. A simple definition of a dietary nutrient requirement is the daily amount of a nutrient necessary to meet a healthy animal’s needs for maintenance, activity, growth, reproduction, and lactation without any change in body reserves or status. That definition implies that the requirement for each nutrient is based on physiological factors and environmental conditions that drive the need for that nutrient. Nutritional needs differ when animals are not in good health but adequate data are generally not available to quantify a “health” requirement; therefore, preservation of good health is considered a component of maintenance. Cows, similar to most mammals, must mobilize body reserves to support lactation during the early postpartum period. Later in lactation, mobilized nutrients must be replenished, and those needs should be considered a requirement.

Conceptually, the term “requirement” suggests that there is a fixed amount of a nutrient required by an animal where no further increase in performance will occur when an animal is fed an additional amount of that nutrient. This principle is the basis for the use of breakpoint analysis to determine a nutrient requirement (Robbins et al., 2006; Pesti et al., 2009). However, animal performance responses to a nutrient seldom follow that pattern. Rather, the typical performance responses to increasing nutrient intake are curvilinear, where increases in animal performance occur at a diminishing rate to increasing nutrient intake (Bath, 1975; Pesti et al., 2009; Liu et al., 2017). In this case, the desired amount of nutrient intake would likely be based on the economic return to an increment in nutrient intake rather than a fixed requirement.

The requirements for individual nutrients provided in previous editions of the Nutrient Requirements of Dairy Cattle actually represent the average responses of individual dairy cattle or groups of dairy cattle that were fed varying amounts

of nutrients. Within any given group of dairy cattle, inherent variability exists in the response of individual animals to a given increment in nutrient intake. Sources of this variation include measurement error, stage of lactation, milk production, body weight, and numerous other differences among individual cattle within a group, as well as true differences among individual cattle that are phenotypically similar. Because of the inherent variance in the requirements among dairy cattle, the use of the term “minimum requirement” has become obsolete.

To overcome some of the problems with the use of minimum requirements, the National Academies of Sciences, Engineering, and Medicine’s Food and Nutrition Board within the Institute of Medicine (IOM)¹ has adopted a new set of nutrient standards (IOM, 2006) collectively referred to as Dietary Reference Intakes (DRIs). In that system, the following terms and definitions are used:

1. The estimated average requirement (EAR) is defined as the average daily nutrient intake estimated to meet the requirements of half of the healthy individuals in a particular life stage and gender group.
2. The Recommended Dietary Allowance (RDA) is the daily dietary nutrient intake sufficient to meet the requirements of nearly all (97 to 98 percent) healthy individuals in a particular life stage and gender group.
3. Adequate Intake (AI) is the average daily nutrient intake that a panel of experts determined should meet or exceed the requirements of a specific group (or groups) based on limited experimental data; AI is used when an RDA cannot be determined.
4. Tolerable Upper Intake Level (UL) is the highest average daily nutrient intake that is likely not to pose a risk of adverse health effects to almost all individuals in the general population.

¹As of March 2016, the Health and Medicine Division continues the consensus studies and convening activities previously carried out by the Institute of Medicine (IOM).

In the DRI system, an RDA can only be determined when the EAR and the variability of the EAR (typically expressed as a coefficient of variation) can be determined. Determination of an EAR requires many feeding experiments in which varying nutrient intakes have been used. The RDA is the EAR plus 2 standard deviations, so that 97 to 98 percent of the individuals in a population that consume the RDA will have sufficient nutrient intake to meet their individual needs. By feeding to meet the needs of 97 to 98 percent of the population, the possibility that an individual will be underfed is almost nil, but the majority of individuals within the population will consume excess nutrient. In the DRI system, the only exception to feeding to meet 97 to 98 percent of the population is for energy requirements, in which even a moderate excess in energy intake over the long term can have severe negative impacts on health such as obesity and type 2 diabetes. For many nutrients, adequate data are available to establish an EAR, but because of factors such as different response measurements and different experimental designs among studies, inadequate data are available to obtain an accurate estimate of the standard deviation. In those situations, the RDA is equal to 1.2 times the EAR because the coefficient of variation in energy metabolism in similar humans is about 20 percent, and variation in metabolism of other nutrients was assumed to be similar to that of energy. In the DRI system, requirements expressed as AI are reserved for situations when an EAR cannot be determined. Typically, this is where insufficient numbers of feeding experiments have been conducted with the target species, and the variability of the response to a nutrient is so great that an EAR and hence the RDA cannot be determined. Finally, the UL is conceptually similar to maximum tolerable levels (MTLs) reported for minerals (NRC, 2005) and vitamins (NRC, 1987) in animal nutrition. However, UL includes an “uncertainty factor” so that the UL is below the level (sometimes much lower) at which an adverse effect may be observed. The MTL is the level at which an adverse (but not necessarily toxic) effect was observed.

The committee adopted an approach similar to the DRI system in establishing the nutrient requirements for dairy cattle. When the term “requirement” is used in this publication, it is equivalent to the EAR used in the DRI system and reserved for nutrients in which the average requirement is known with confidence. When possible, measures of variation were included in the text. The variation might be determined among treatment means, among animals within studies, or from a meta-analysis or by regression analysis where the mean predicted response and the standard error of the estimate of the predicted response have been determined. Similarly, in this report, when a suggested feeding amount of a nutrient is expressed as an “Adequate Intake,” it means that insufficient data have been collected to determine an estimated average requirement. For some of the vitamins and trace elements, there are currently insufficient data available to determine an average requirement. In many cases, this is due to the wide variability in nutrient availability or the lack of sufficient stud-

ies with multiple feeding levels to determine an EAR and its variance. The committee does not specify an equivalent to an RDA (i.e., a safety factor of 2 standard deviations) for dairy cattle because of limited data and because of the economics of dairy production. For several nutrients, inadequate data are available to estimate the standard deviation of the response to nutrient supply. Furthermore, meeting the equivalent of an RDA (i.e., requirement plus 2 standard deviations) may cause the diet to exceed the MTL for some nutrients.

Feed is the largest single expense in raising and caring for dairy cattle. The ingredients used and the nutrient composition of diets have large effects on the economics of dairy production. For some nutrients that are relatively inexpensive to supplement, such as certain vitamins and trace minerals, the cost of feeding to meet 97 to 98 percent of the animals in a group would be low. However, for macronutrients such as energy, protein, and some of the macrominerals, the cost of such an approach would be high. Depending on the nutrient, feeding sufficient amounts to cover 97 to 98 percent of the cows within a group would likely be uneconomical, may cause environmental issues by excess excretion of the nutrient in the manure, and, in the case of energy, result in overconditioned cows.

REPORTING AND APPLICATION

OF THE REQUIREMENTS

The majority of the requirements in this publication are reported as absorbed (minerals) or metabolizable (energy, protein, and amino acids) nutrient intakes. Examples include metabolizable energy, metabolizable protein and amino acids, and absorbed minerals. When nutrients are expressed this way, a reliable means to estimate nutrient supply in the same terms is required. For some feed ingredients and nutrient classes, the ability to predict nutrient availability is inadequate, and this is discussed in the chapters on individual nutrients.

Similar to previous reports (NRC, 1987, 1989, 2001), a factorial system has been used to express the requirements for most nutrients according to physiological function and the amount and composition of production. This is discussed in detail in various chapters, but using energy requirements as an example, maintenance energy requirements are based on an animal's body weight and include a fixed adjustment assumed to account for normal activity for cattle that are not grazing. An activity allowance based on topography and distance to the milking center is included if cows are grazing. Energy requirements for growth are based on an animal's growth rate and the composition of growth. Energy requirements for reproduction are based on the stage of gestation and size of the fetus and uterus. Finally, energy requirements for milk production are based on the amounts and composition of the milk produced. The animal's total energy requirements are the sum of the individual requirements for maintenance, growth, reproduction, and milk production.

By inclusion of both the average requirement and, when possible, measures of its variability, the committee has laid the groundwork for incorporating some of the principles identified in the human DRI system. This variability may or may not have been captured if the average requirement had been based on group means or pen feeding experiments reported in the literature as compared to using individual animal data. Even when experimental treatments were applied to individual animals, the experimental design impacts reported variation. Reported variation is greater from experiments using completely randomized designs as compared to designs in which individual animal variation is removed (e.g., Latin square designs or designs with covariance in the statistical model). This has the greatest impact when an index of variance is used in weighting study effects in the meta-analysis to determine the response to a nutrient. In addition, reported variation includes not only true animal-to-animal variation but also measurement variation, which for some responses can be quite high.

In addition to variation in requirements among individual animals, uncertainty with respect to diet composition needs to be considered before an RDA approach can be used. Recent publications have documented the uncertainty in the knowledge of feed ingredient composition. Some of the uncertainty in feed composition is due to the mislabeling of feeds being submitted for analysis to feed analytical laboratories that were used as a source of data for feed composition tables (see discussion in Chapter 19). The degree of variation in nutrient composition varies greatly among feed ingredients; some ingredients are consistent enough that sampling is not required, whereas composition of other feeds is so variable that frequent sampling is needed to ensure that the nutrient content of the diet can be verified (St-Pierre and Weiss, 2015). Day-to-day variation was the greatest source of variability in diet dry matter concentrations, whereas individual farm, month-to-month, and sampling was the greatest source of variation for other nutrients (St-Pierre and Weiss,

2015). Because of the expense of feed analysis, strategies for ingredient sampling and analysis have been proposed (St-Pierre and Cobanov, 2007) such that feeds and nutrients within feeds that have high inherent variation in composition are analyzed more frequently.

Estimates of variability in nutrient requirements and feed composition could be incorporated into multiobjective diet formulation procedures in the future so that users can set a specific probability that dietary nutrient constraints (requirements) are met (i.e., stochastic formulation). Diet formulation procedures based on the cost and uncertainty of ingredient nutrient composition have been identified (St-Pierre and Harvey, 1986 a,b; Tozier and Stokes, 2001). In general, these approaches result in greater numbers of individual feed

ingredients and selection of feeds with lower inherent variability in nutrient composition being incorporated into the diet (St-Pierre and Harvey, 1986a,b) and often increase the cost of the diet depending on the risk (or probability of meeting requirements) one is willing to accept. Some of the increased cost could be balanced against increased production because of reduced variability in diet nutrient concentrations. The optimal amounts of nutrients to be fed depend on the production responses to nutrient intakes in relation to their cost, the uncertainty in the knowledge of the actual nutrient concentrations of the feeds within a diet, and the variability in the requirements among dairy cattle within the group (St-Pierre and Harvey, 1986c; Cabrera and Kalantari, 2016).

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